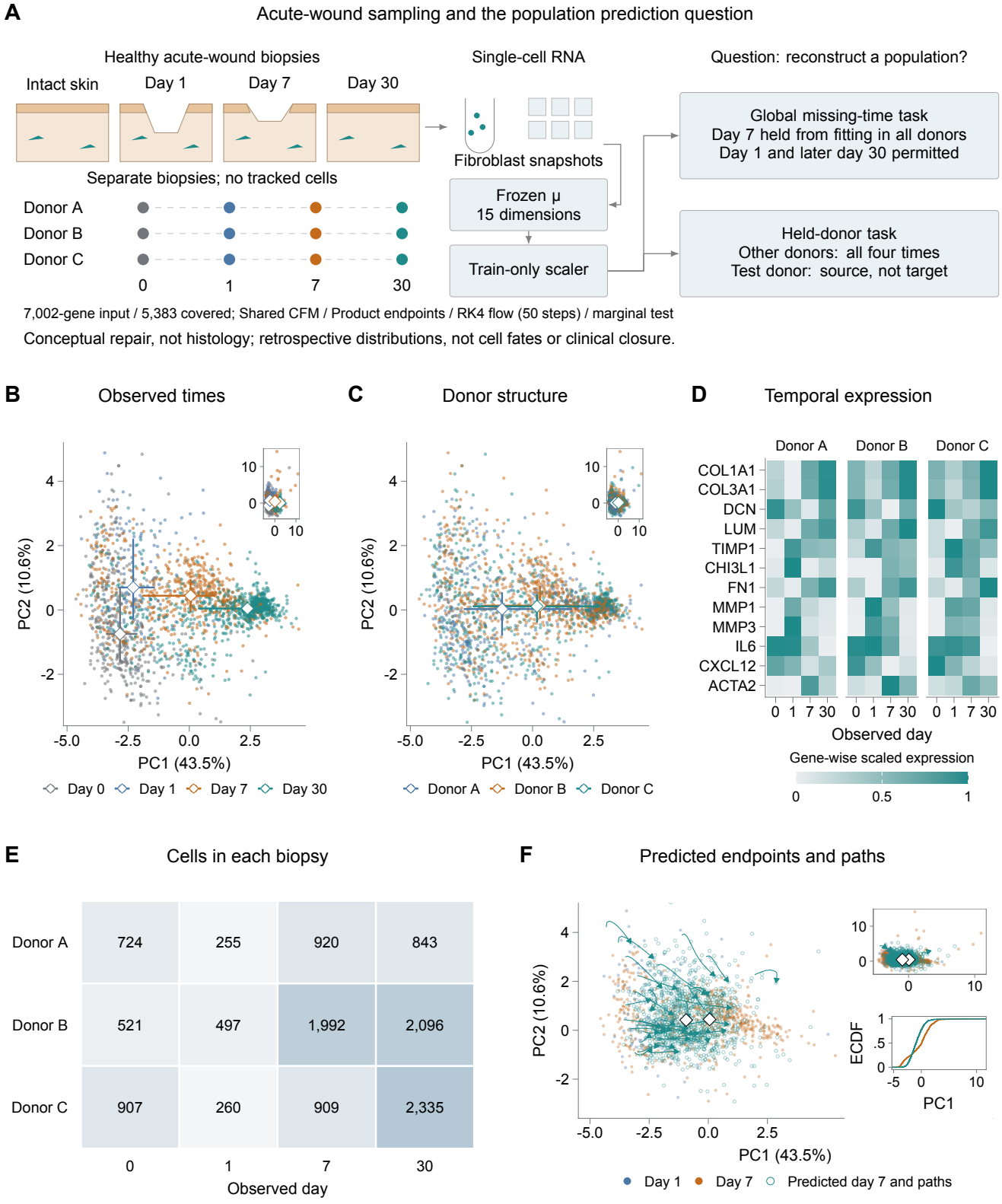
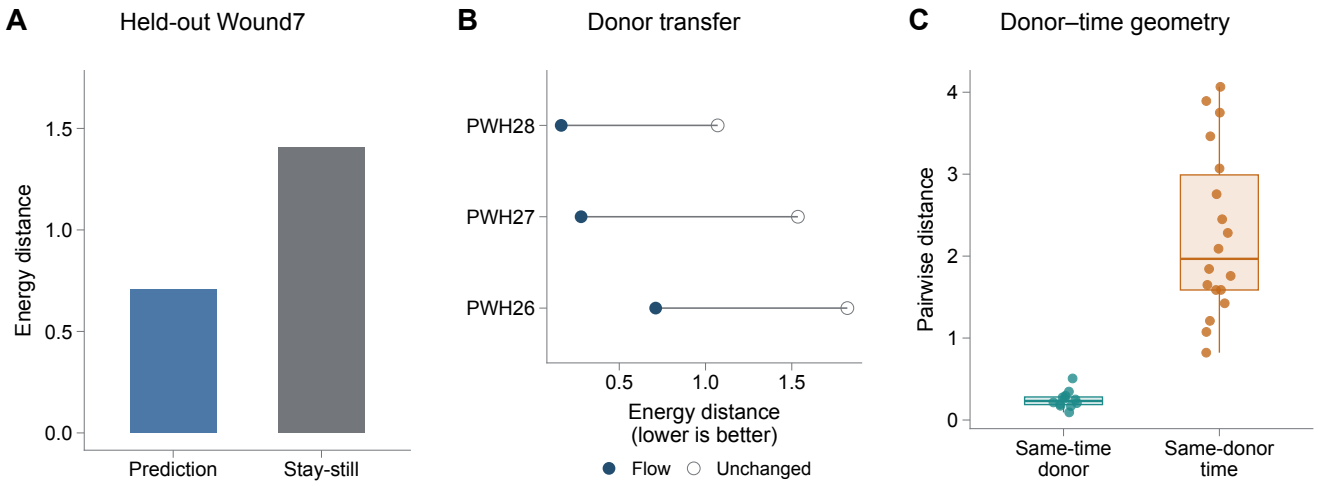
Figure 1



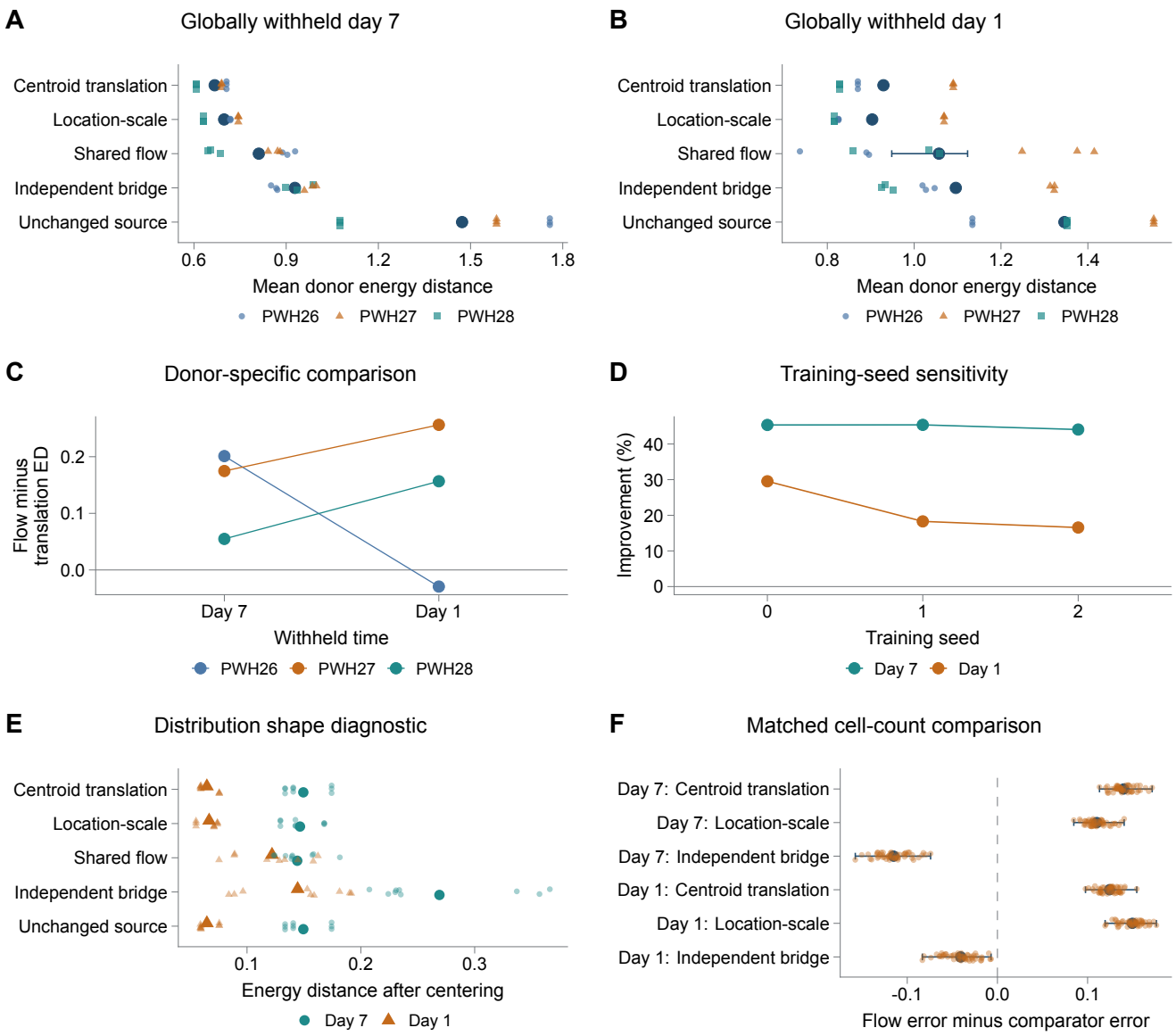
What can acute-wound snapshots predict? (A) Conceptual repair and separate biopsies from three healthy donors at days 0, 1, 7 and 30; cells are not tracked. Global missing-time reconstruction excludes day 7 in every donor and permits later anchors. Held-donor transfer permits training-donor day-7 cells but excludes the test donor's target. Neither is clinical prognosis. A 7,002-gene input with 5,383 covered genes feeds frozen 15-dimensional μ, training-only scaling, product-endpoint CFM and 50-step RK4. Flow acts in scaled μ coordinates, not simplex weights. (B) Saved 2,000-cell display by time; day-7-excluded PCA explains 43.5% and 10.6% of training variance. (C) The same cells by donor. Diamonds/crosses denote medians/interquartile spans, not confidence intervals; focus windows retain full-range insets. (D) Twelve measured genes per biopsy, gene-wise scaled after pseudobulk normalization. (E) Recovered fibroblasts, not independent donors. (F) All 1,012 saved seed-0 endpoints, 36 numerical paths and observed day-1/day-7 display; insets show full range and PC1 cumulative distributions. Paths are not cell fates; errors use all 15 coordinates.

Figure 2



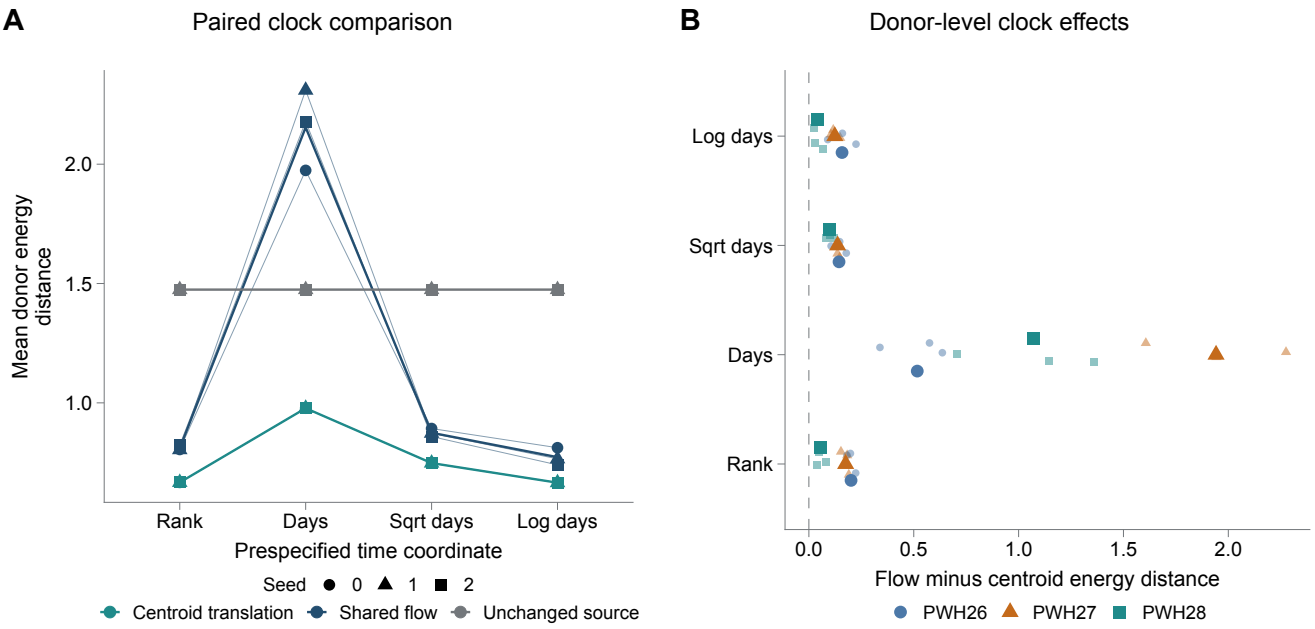
Global Wound7 reconstruction, donor transfer and donor–time geometry. (A) Predicted and unchanged-source energy distances to pooled Wound7 after excluding that time globally; relative improvement is 49.9%. (B) Connected filled and open points give predicted and unchanged-source errors for each genuinely excluded donor; the other two donors retain all four times. Mean relative improvement is 75.9%, but absolute baseline and prediction errors differ by donor. These are point estimates, not confidence intervals. (C) Twelve same-time between-donor and 18 same-donor between-time distances, with interquartile boxes, medians and whiskers to observations within 1.5 interquartile ranges. The median temporal/donor ratio is 8.46; its 10,000-draw pair-distance resampling range is descriptive. All distances use training-standardized latent coordinates. Shared donors make pairs dependent, so these ranges have no established population coverage; three healthy donors do not validate DFU prediction.

Figure 3



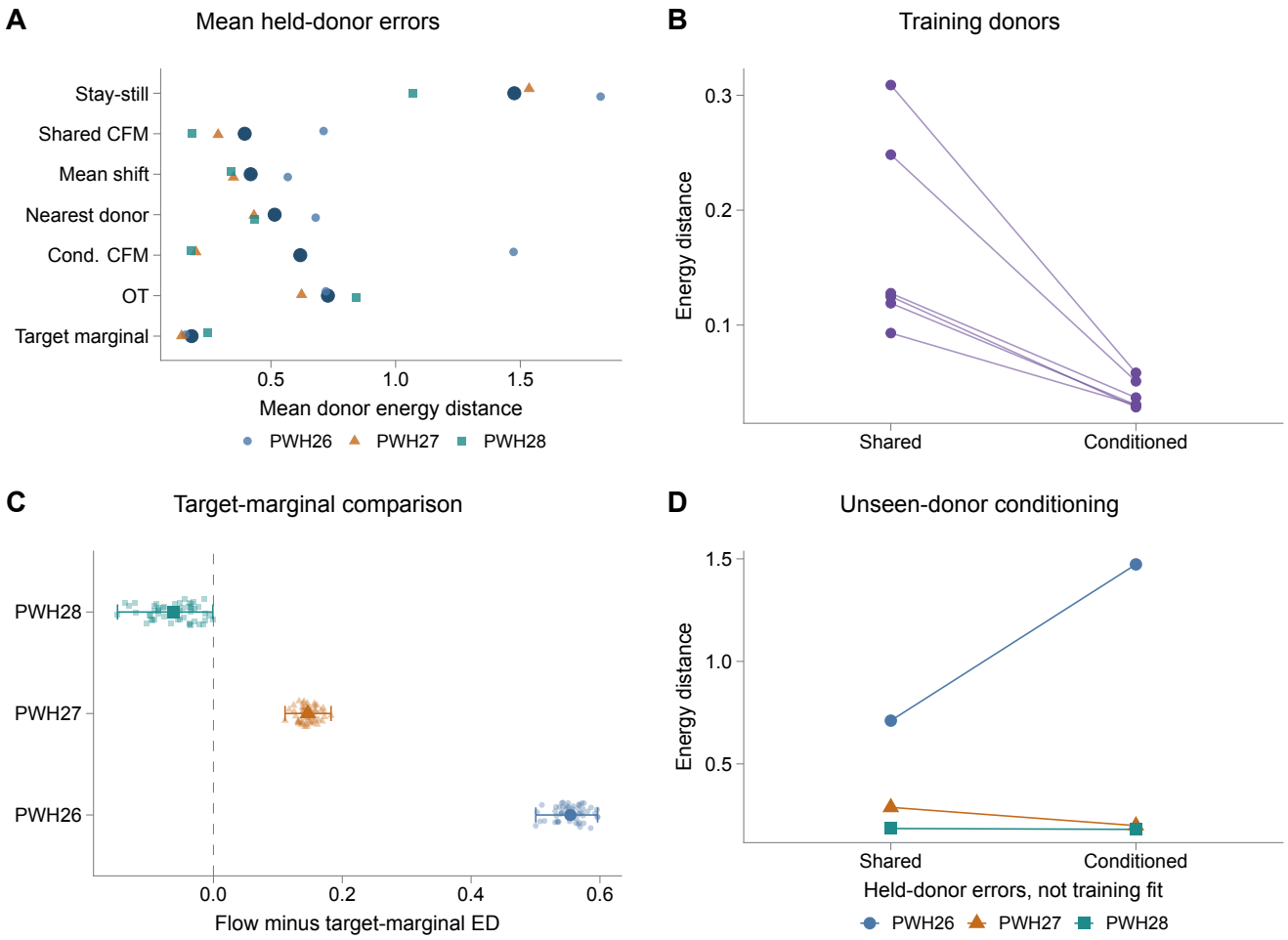
Simple baselines for globally missing times. (A) Day 7 from day 1 with observed day 30 as anchor. (B) Day 1 from intact skin with observed day 7 as anchor. Large points are donor-macro means and bars span three seeds; smaller colored symbols retain individual donor/seed errors for all five predictors, including location–scale. Both tasks use rank-time fraction one half and are retrospective interpolation, not source-only prediction. (C) Donor-specific seed-averaged flow-minus-translation error. (D) Each flow seed's relative improvement. (E) Seed/donor-averaged centered energy distance with the individual donor/seed errors behind it; this is a shape diagnostic rather than additive error decomposition. (F) All 50 saved donor/seed-averaged paired draw contrasts, overlaid with means and ranges, using 200 source and target cells per donor. Positive differences favor the comparator. Seeds and draws are dependent computational evaluations of three donors; neither their ranges nor their point clouds are biological confidence intervals.

Figure 4



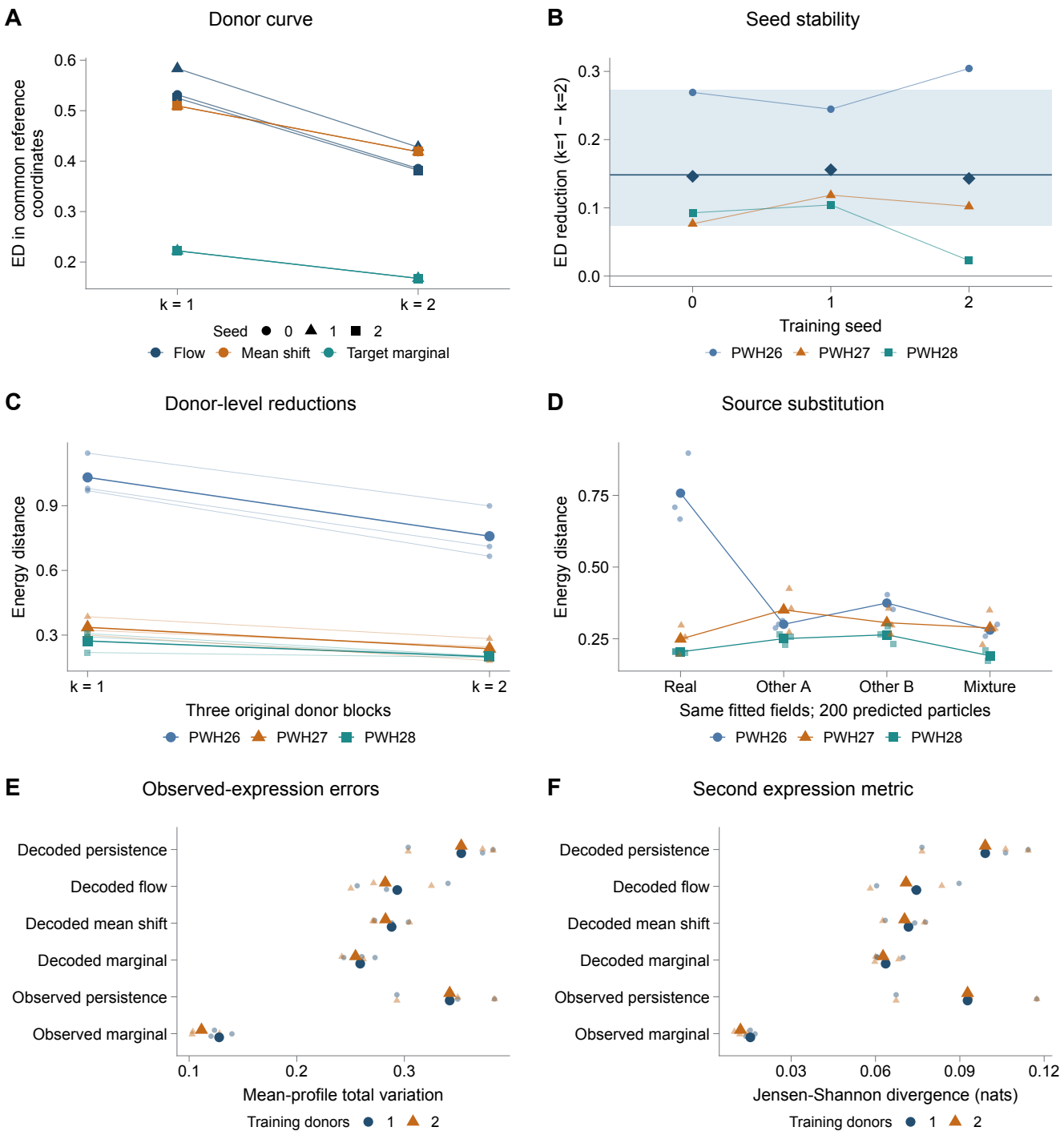
Paired time-coordinate sensitivity under global day-7 holdout in three healthy donors. Twelve fields cross four clocks with seeds 0, 1 and 2, each using 8,000 training steps and batch size 256. Initialization, endpoint draws and bridge random draws are paired across clocks within seed. Day 7 is excluded from fitting and scaling; 50 RK4 steps reach its exact transformed time, with identical evaluation rows and a 1,200-cell cap per donor. (A) Points are equal-donor means, shapes identify seeds, thin lines connect each seed and thick lines show seed means. Colors distinguish flow, centroid translation and unchanged source. Sqrt denotes square-root days; constant baselines overlap. (B) Large points show each donor's three-seed mean flow-minus-centroid error; small points retain the three seed-specific errors. Positive values favor the centroid and the dashed line marks no difference. Colors and shapes identify PWH26, PWH27 and PWH28 consistently across result figures. The centroid uses permitted day-1 and day-30 biopsies, so this remains retrospective interpolation. Clock connections are not biological trajectories; seed spread is not population uncertainty.

Figure 5



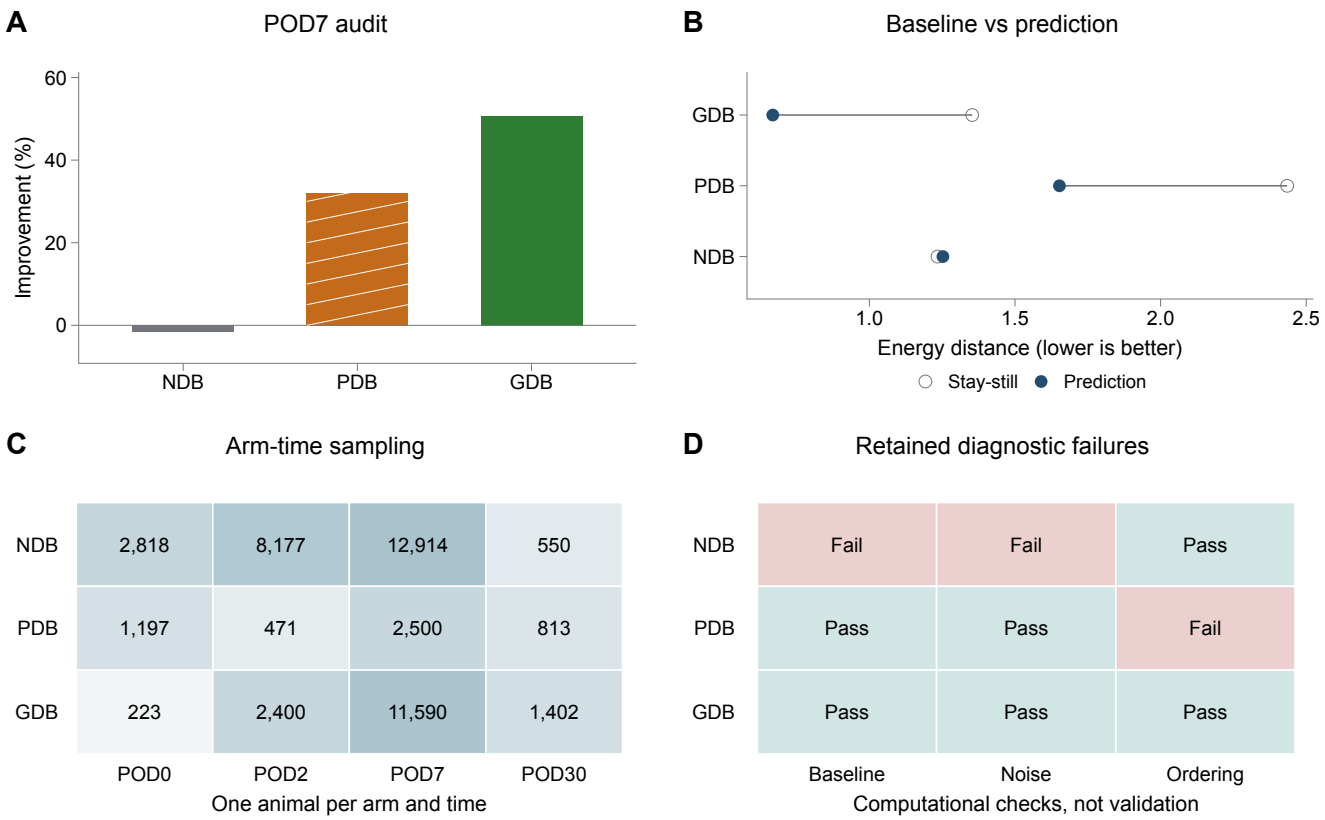
Method means and donor-specific reversals in the historical held-donor benchmark. (A) Navy points are equally donor-weighted mean energy distances for unchanged source, five predictors and the added target marginal; smaller colored points retain all three donor errors for each method. CFM is conditional flow matching; OT is the separate entropic transport comparator. The marginal assigns equal mass to training-donor day-7 distributions without held-source expression; 50 matched-particle draws use the original target rows and scaling. (B) Six shared-versus-conditioned comparisons on training donors, all favoring conditioning; these are overlapping training observations, not transfer tests. (C) Each donor's fixed shared-field error minus the marginal's mean error, with the full 50-draw range and all 50 saved contrasts as smaller points. Positive values favor the marginal; PWH28 instead favors the field. These ranges describe prediction sampling, not donor-population confidence. (D) Connected shared and conditioned errors for the three genuinely held donors, separately exposing the gap between training fit and unseen-donor behavior. Historical neural/transport predictors used one training seed; the retrospective marginal audit does not establish a stable method ranking.

Figure 6



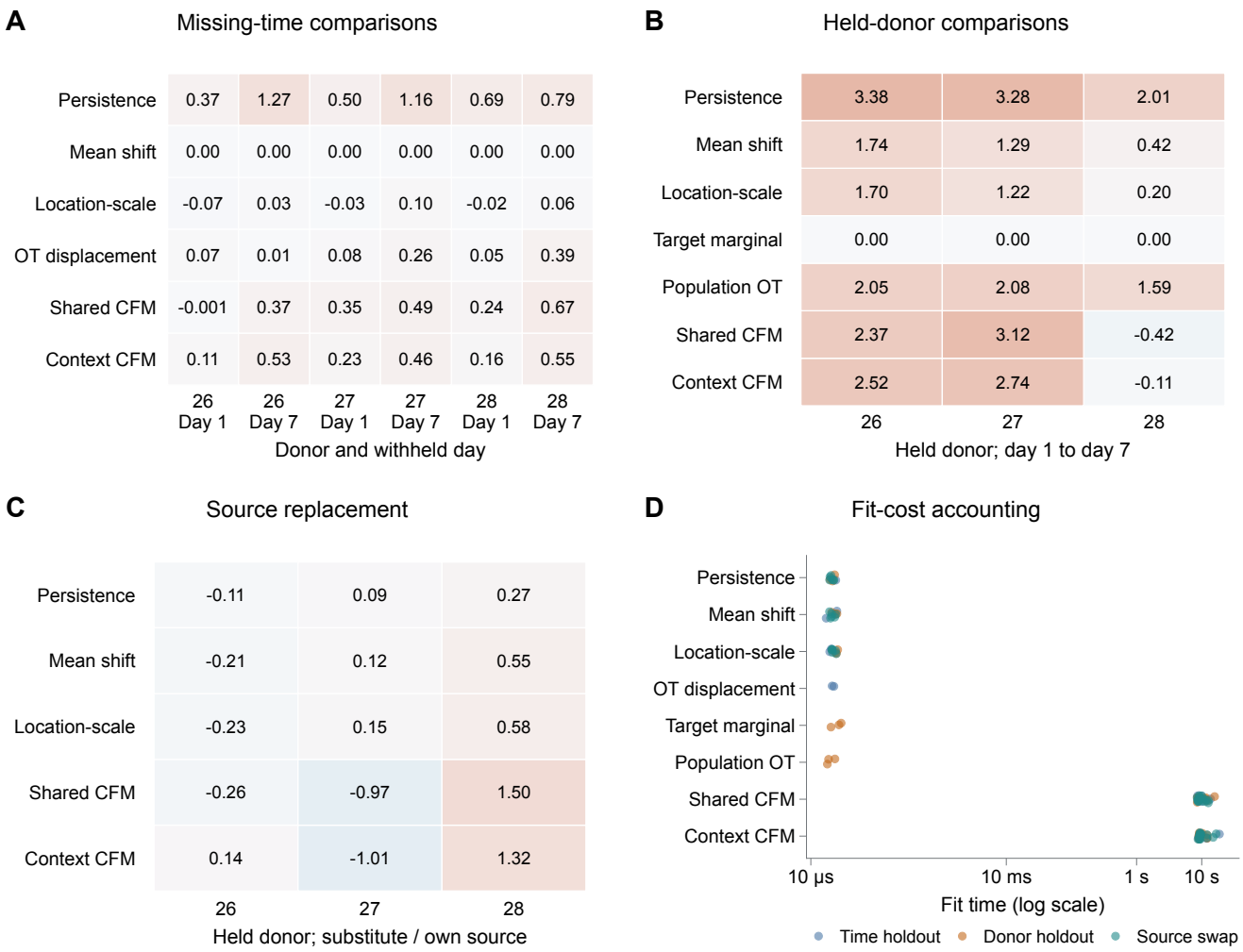
Donor count, source substitution and observed-expression restrictions. (A) Three-seed shared-flow curves alongside saved mean-displacement and exact target-marginal curves, in the same reference geometry and hierarchical donor weighting. (B) Macro seed reductions (navy diamonds; navy line, their mean), all nine donor/seed reductions and the saved descriptive three-block 95% range; shading is not a seed-level interval. (C) Donor means with finer seed-resolved curves. (D) Real source, each alternative donor and their mixture under the saved 200-particle, 50-draw protocol; large points are donor means and smaller points are the three seed means. Other A/B are sorted alternatives within each fold. (E) Total variation against observed mean expression on 5,383 common genes. (F) Jensen–Shannon divergence for the same predictions. E/F distinguish decoded and directly observed predictors; large symbols are equal-donor macros and faint symbols are three donor means after seed/subset averaging, for each training size. Source mixing is not a donor-count experiment, and mean-expression errors are not full cellular-distribution validation. All panels reuse three donors; seeds, draws and overlapping folds add no biological replication.

Figure 7



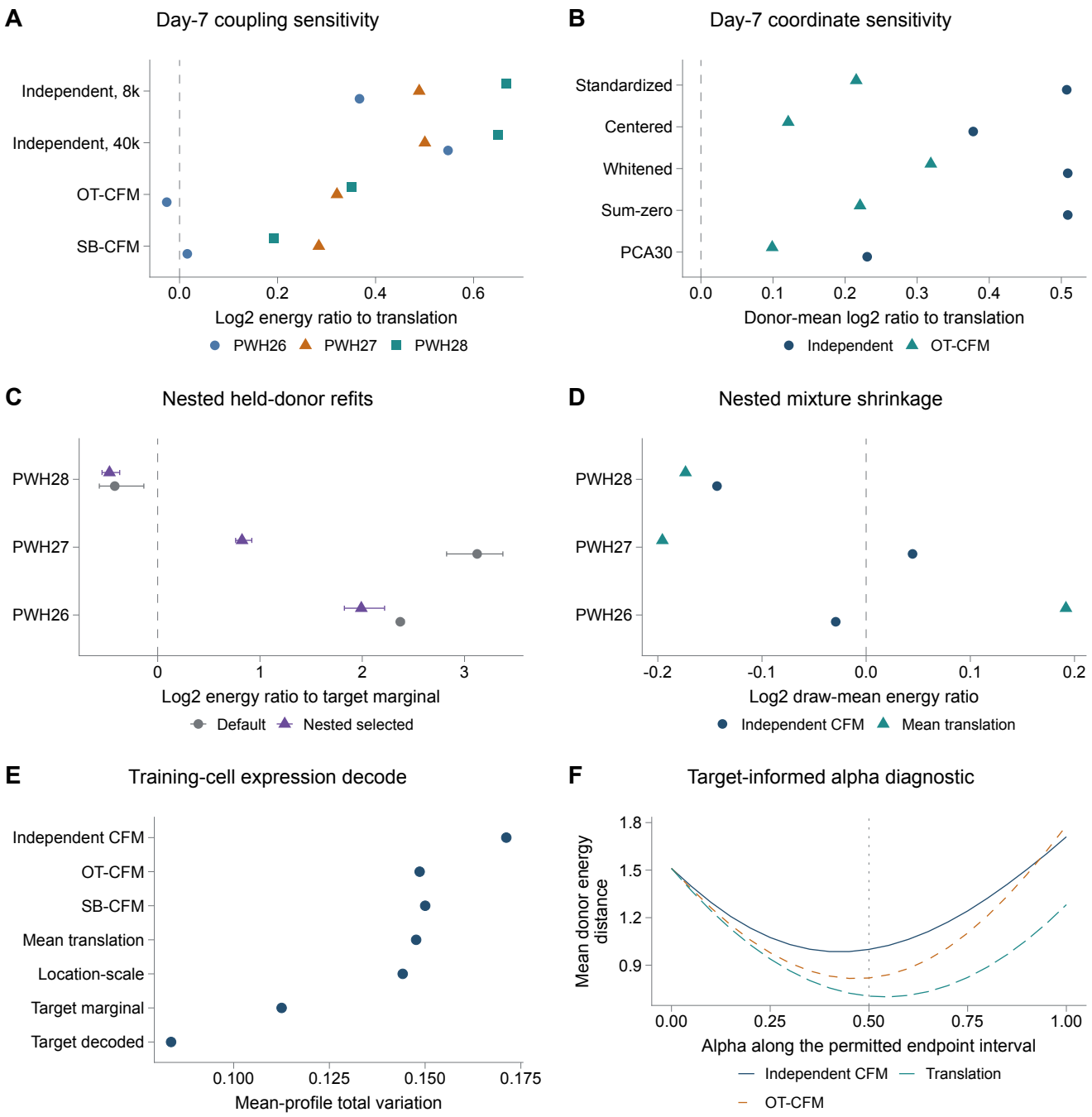
Mouse POD7 computational audit for NDB, PDB and GDB source arms. POD7 is excluded from fitting and scaling; fields trained at POD0, POD2 and POD30 predict from POD2. (A) Relative improvements −1.5%, +32.1% and +50.6%; PDB hatching marks the failed endpoint-ordering check. (B) Connected unchanged-source and prediction errors on the absolute scale. (C) All 12 saved arm-time cell counts; color follows log-count and printed values are exact counts. Each entry has one animal, not multiple biological replicates. (D) Separate baseline, split-half noise and endpoint-ordering check outcomes, preserving every failure. Neither pass labels nor endpoint ordering establish biological trajectories. All arms cover 4,182/7,002 panel genes; no biological confidence intervals are available, and cells are not independent animals or clinical validation.

Figure 8



Source-bound repeated comparisons in the frozen representation. (A) Each cell is the mean paired log2 energy-error ratio to mean translation within a donor and withheld day, averaging 50 matched draws within a fit and then three neural training seeds. (B) Day-1-to-day-7 donor-transfer ratios use the permitted equal-training-donor target marginal instead. Negative values favor the row method; zero denotes its comparator. Neither display pools raw distances from differently standardized donor folds. (C) Source replacement divides each substituted-source error by the own-source error from the same field and target draw, then averages draws, seeds and the two substitute choices within donor. These separate fits exclude both the held donor and the substitute from fitting; positive values indicate lower error from the own source. (D) Every one of the 107 fitted-job timings is counted once, not once per evaluation draw; colors distinguish missing-time, held-donor and source-replacement tasks. Fit time excludes representation setup, prediction, scoring and input/output and is not an equal-compute comparison. Original-CUDA-backend replay and the failed CPU portability check are distinguished. All panels reuse three healthy acute-wound donors, without inferential method ranking or clinical validation.

Figure 9



Completed computational follow-up. (A) Missing-day-7 log2 energy ratios to translation in each donor, across coupling families and two independent-field budgets. (B) Donor-mean day-7 ratios in five training-fitted coordinate systems. (C) Default and nested selected day-1-to-day-7 fields relative to the training target marginal; whiskers span three seed means. (D) Nested mixtures of transported source and target marginal, using log2 ratios of draw-averaged energies, a different estimand from A–C. (E) Held-donor mean-expression errors after mapping predictions to 15 nearest permitted training cells; decoding observed target cells supplies a reference. (F) Target-informed alpha scans of fixed rank-trained predictors, with rank alpha 0.5 dotted. Alpha optima are retrospective diagnostics. All panels reuse three human donors; draws and seeds add no biological replication.